UNIVERSITY OF WATERLOO

MTE 203
Advanced Calculus

Spring 2026

Welcome to MTE 203 Spring Term 2026

Instructor: Prof. Patricia Nieva

- **Email**: pnieva@uwaterloo.ca
- **Office hours**: Mondays 4:30pm – 5:30pm, E5-3052 (Starting May 25, 2026)
- **For questions of a personal nature,** please email me.

Teaching Assistants

- **Nahvid Zolfaghari** (n4zolfag@uwaterloo.ca)
  - Office hours: Wednesdays 4:30pm – 5:30pm, Room E5-3048
  - Starting on May 20, 2026
  - Email outside office hours
- **Jun Wang** (j872wang@uwaterloo.ca)
  - Matlab related questions
    - Lab 4 Help Session: Friday Jun 17, 2026, 4:30pm – 5:30pm, Room E5-3048
    - Contact via email during the term

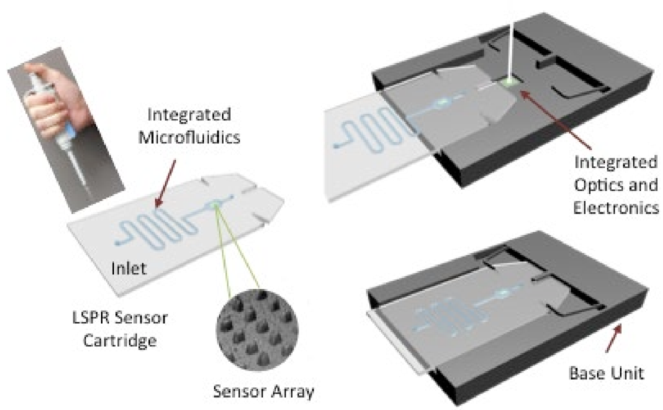
UNIVERSITY OF WATERLOO

About me

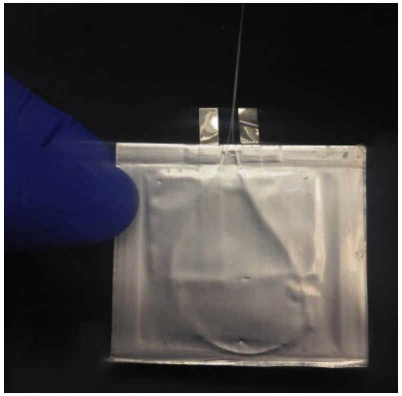
SiμS Laboratory
SENSORS AND INTEGRATED MICROSYSTEMS

My core research thrust deals with the development of smart integrated microsensor and nanosensor solutions for harsh environments, automotive applications, health monitoring and medical diagnosis.

COVID-19 Intelligent Alert System using Bluetooth-based wearables.



BioMEMS for Cortisol Sensing



In-line Sensors for Monitoring of State of Charge (SOC) in Li-Ion Battery Cells

UNIVERSITY OF WATERLOO

If you are ill

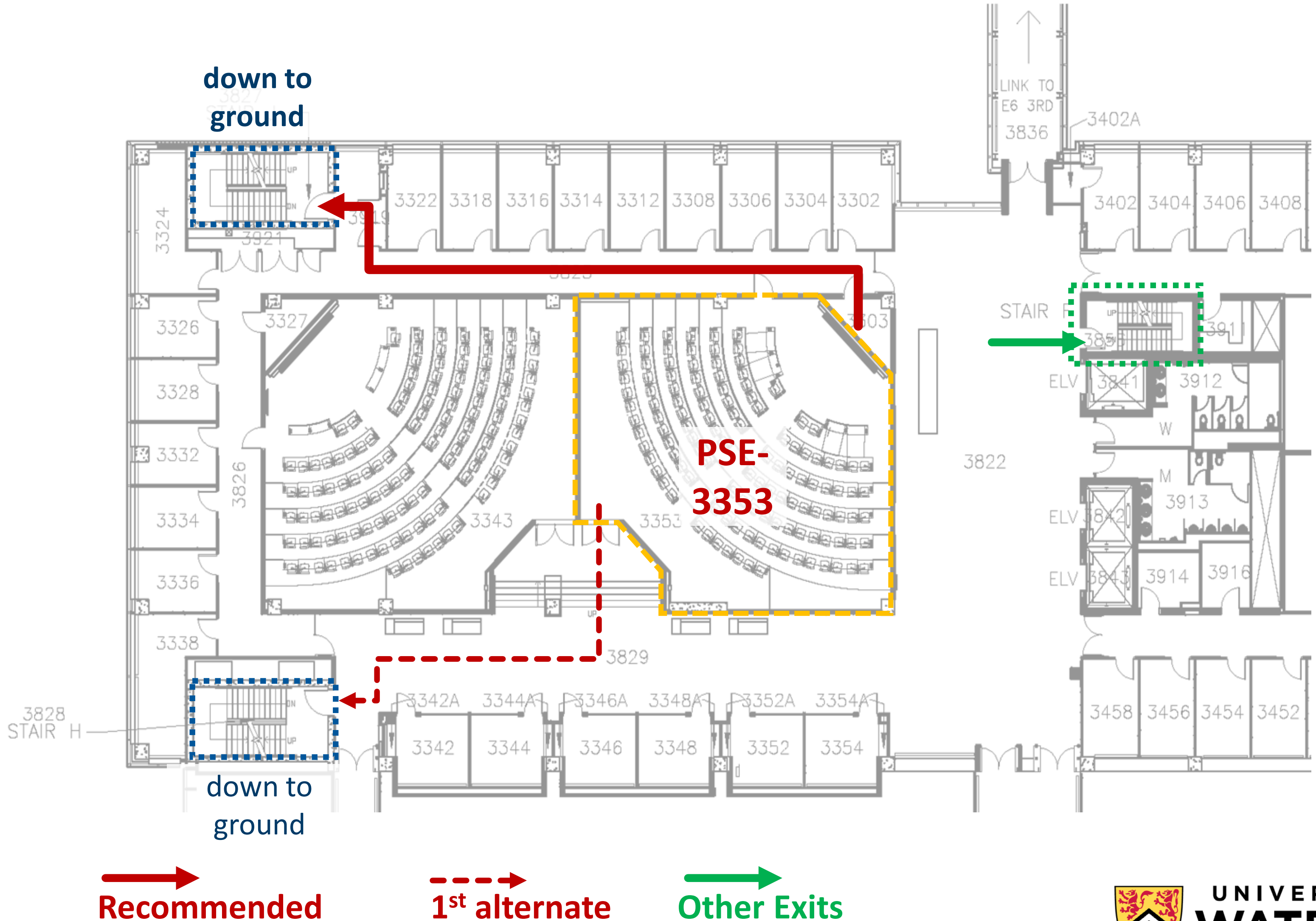
- DO NOT attend class
- If you're on campus, go home immediately
- Stay home until symptoms have been improving for 48 hours, you have no fever, and you haven't developed any new symptoms
- Self-Declare on Quest (for short-term absences up to two days) or use the VIF system

CLASSROOM EVACUATION PROCEDURES

If the **fire alarm** sounds **or** you are **directed to evacuate**:

- Calmly evacuate the building through your closest emergency exit.
- Do not use the elevators.
- If time allows close windows and doors. If you encounter smoke or fire, use an alternate exit

PSE-3353 Recommended Emergency Exits



UNIVERSITY OF WATERLOO

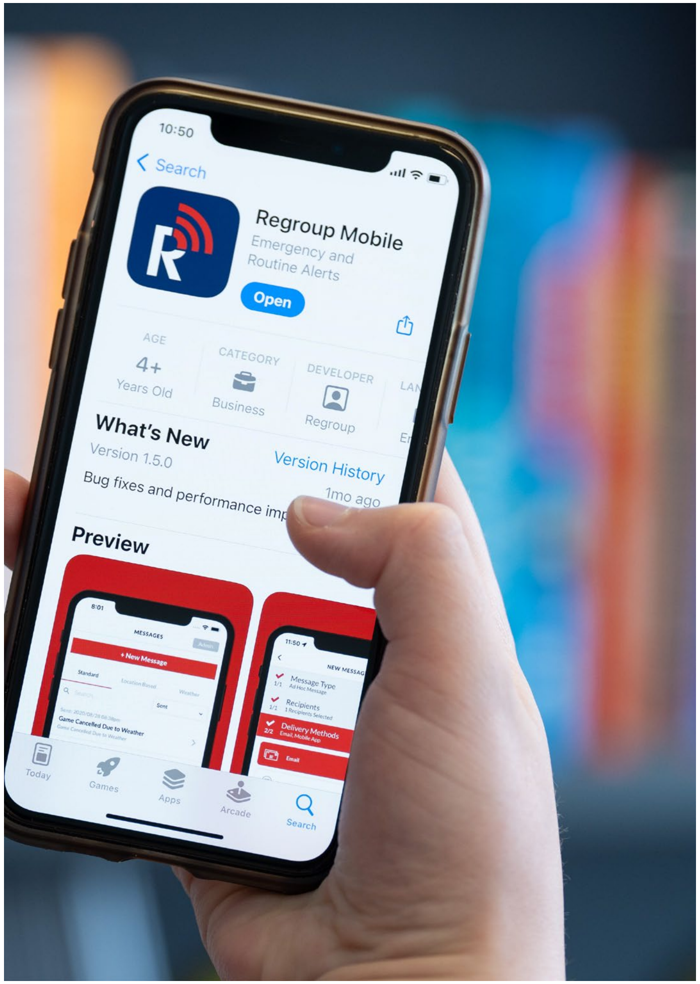
DOWNLOAD REGROUP MOBILE NOW

Download the app!

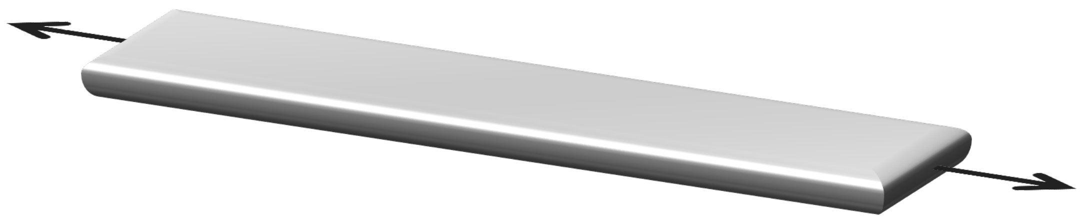
[Go to the Regroup Mobile Website]

Need support?

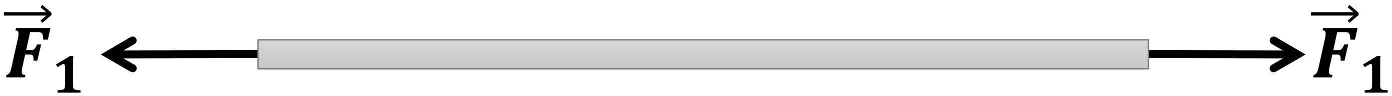
Contact [Counselling Services]

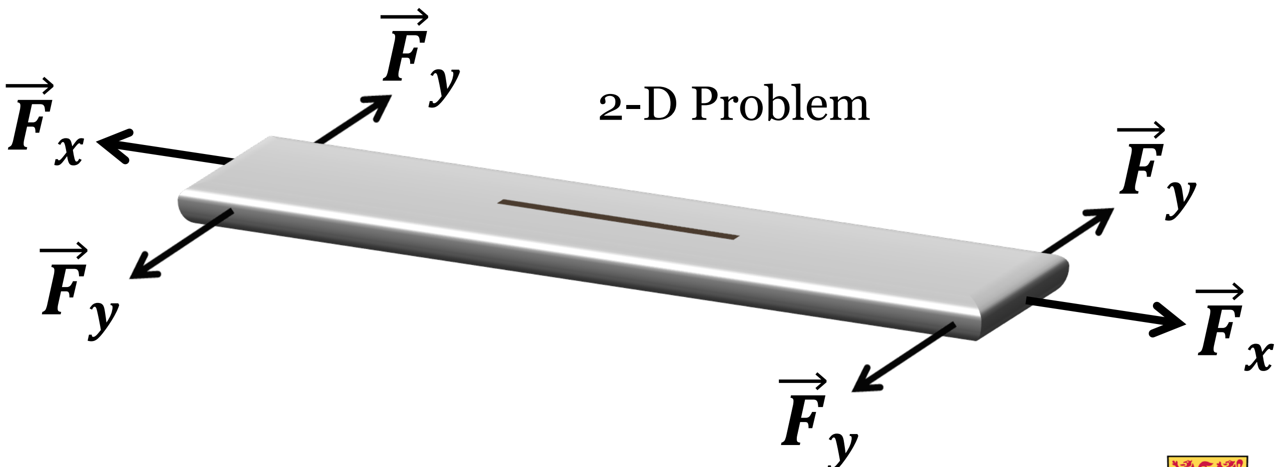


Why is 2-D & 3-D Math important?



1-D Problem





Calculus and Hurricanes?

Hurricane Melissa
Hit Jamaica as a Category 5 on October 28, 2025

Hurricane's eye diameter is between 20-50km. Melissa's eye ranged between 8-21km, characteristic of its extreme, high-end intensity

Major Hurricane Melissa makes landfall near New Hope, Jamaica on October 28, 2025

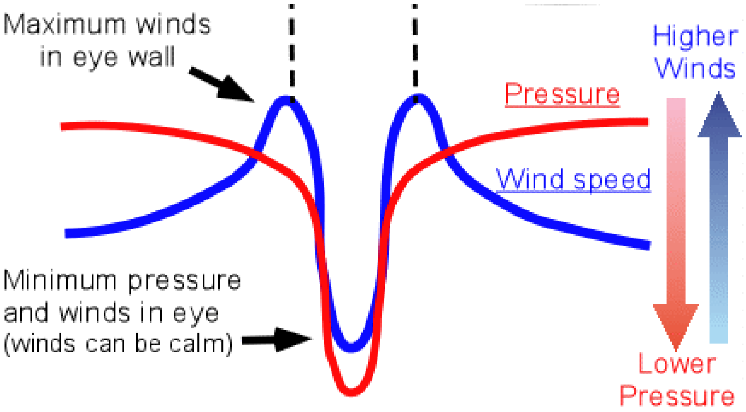
https://satlib.cira.colostate.edu/event/hurricane-melissa/

How to safely navigate close to the storm?

The "Hurricane Hunters," flew multiple passes through Melissa on Monday October 25, 2025, to collect critical weather data for the National Hurricane Center.

https://www.youtube.com/watch?v=O_LIuonLTiQ

First of two unmanned Global Hawk aircrafts!



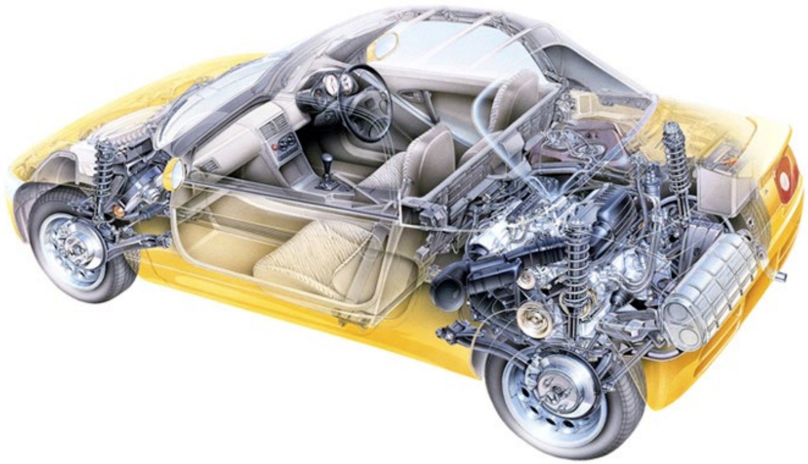
UNIVERSITY OF WATERLOO

Multiple Integrals and Automobile Handling

Subaru BRZ

Key to Sports Car Handling:
Low Weight and Ultra-Low Center of Gravity
(i.e. point where all the weight could be concentrated – low is better stability)

Polar Moment of Inertia refers to how difficult it is to get an object to rotate on an axis

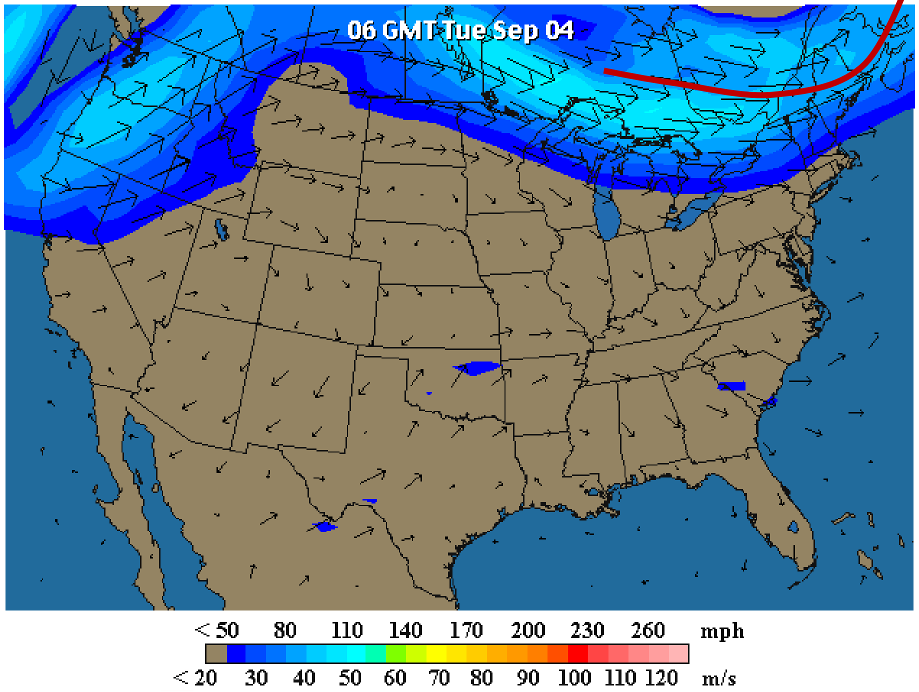


The Honda Beat design allows a very good weight distribution with low polar moment of inertia

Vector Fields and Jet Streams

Weather Forecast Jet Stream

Arrows show wind direction



- The jet stream is like a river of air in the upper atmosphere
- Weather patterns are influenced by the position, strength and orientation of the jet stream.

https://earth.nullschool.net/

Course Objectives

The objective of this course is the mathematical description of phenomena that vary in 2 and 3 dimensions including:

- Scalar functions
- Vector functions
- Derivatives
- Integration

Learning Outcomes

By the end of this course, you should be able to:

- Understand the basic concepts and know the basic techniques of differential and integral calculus of different functions of several variables
- Apply the course concepts to physical systems, for example describing position, velocity and acceleration in 3D space
- Be comfortable with the notation used for multivariable calculus, e.g. ∂, ∇, $\oint$
- Be able to use MatLab to solve engineering problems requiring multivariable calculus

UNIVERSITY OF WATERLOO

Course Website

Hosted through UW-LEARN

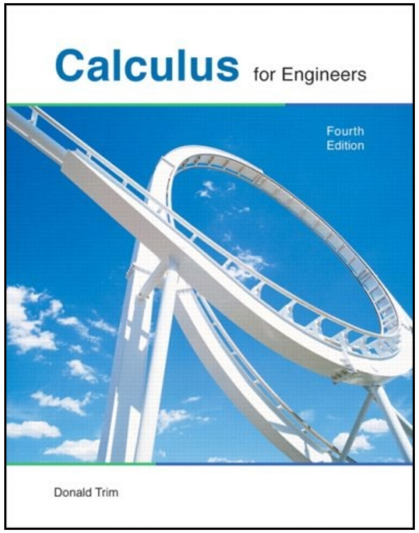
http://learn.uwaterloo.ca

Check the **Course Information LEARN Module** for all information related to the course

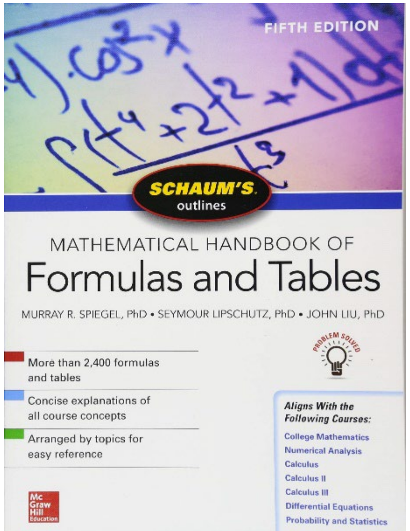
Course Outline available at https://outline.uwaterloo.ca/viewer/view/nwcn94

UNIVERSITY OF WATERLOO

Textbooks



Calculus for Engineers, Donald Trim, 4^{th} Edition and Student Solution Manual
Required



Mathematical Handbook of Formulas and Tables, 5^{th} Edition M. R. Spiegel, Schaum's Outline Series
Recommended

UNIVERSITY OF WATERLOO

Grade Breakdown

Activities and Assignments	Weight (%)
Self-Assessment Quiz	2%
Syllabus Review Quiz	Ungraded[1]
Knowledge Review Quiz 1	Ungraded[2]
Knowledge Review Quizzes (10 at 1% each)	10%[2]
Matlab Worksheets (4 at 1%, 2%, 2% and 5%)	10%
Midterm Exam (Weeks 1-6) - 2 hours	30%
Final Exam (Weeks 8-13) - 2.5 hours	48%
I-clicker online - BONUS	5%[3]

[1] Must attain 100% to gain access to the knowledge review quizzes

[2] Must attempt/submit to unlock next week's Knowledge Review Quiz.

[3] A 5% MAXIMUM BONUS will be assigned for at least 75% of i-clicker questions answered (see details in the i-Clicker Cloud Information for MTE 203 Spring 2026 section in LEARN).

Self- Assessment Quiz [2%] (Due May 12, 11:59PM)

- Vectors and Matrices
  - A vector refresher is posted in UW LEARN (Weekly Materials – Week 1) [Ch11, Sec.1.1,1.3,1.4,1.6]
- Calculus in the Plane (R^2)

Basic Calculus – Chapter 1

- ❑ Straight lines [Ch1, Sec.1.3]
- ❑ Conic sections [Ch1, Sec. 1.4]
- ❑ Graphing of R and R^2 functions [Ch1, Sec.1.5]
- ❑ Trigonometry Review (Ch1, Sec 1.7)

Differentiation – Chapter 3

- ❑ Methods of Differentiation [Ch3, Sec. 3.1, 3.2, 3.4,3.5, 3.7]

Techniques of Integration – Chapter 8

- ❑ Methods of Integration [Ch.8, Sec. 8.1, 8.3, 8.4]

Parametric Equations and Polar coordinates – Chapter 9

- ❑ Parametric equations [Ch9, Sec. 9.1]
- ❑ Polar coordinates and their curves [Ch9, Sec. 9.2 and 9.3]

UNIVERSITY OF WATERLOO

Assignments and Tutorials

- **Weekly assignments and their Solutions**
  - Posted on Wednesdays for the current week
  - Delivered by the TA or the Instructor
  - Instructor tutorials are announced in advance using LEARN and may use the i-clicker polling system
- **Eleven tutorials**
  - Questions posted the same day as the homework
  - Solutions posted immediately after the tutorial session
  - No tutorial during Midterm Exam Week and Canada Day
  - **Highly recommended** as it complements the class material
- **First tutorial: May 13 – Instructor**

Knowledge Review Quizzes

- **Eleven Knowledge Review Quizzes [10%]**
  - Available each week on Fridays at 12:30 PM and due the following Wednesday at 11:59 PM
  - Week 1 Knowledge Review Quiz is Ungraded
  - Lowest Knowledge Review Quiz grade is dropped
  - Week 11 Knowledge Review Quiz is optional and replaces second lowest Quiz Grade
  - The knowledge review quiz for the following week will be available to you only if you submit/attempt the knowledge review Quiz for the previous week.
  - Missed knowledge Quizzes count for ZERO points in the corresponding Quiz (No MAKE UPs)
- Check the Assessment and Activities in the Course Information Module in LEARN for all deadlines and assessment weights listed per week.

UNIVERSITY OF WATERLOO

Four Mandatory MatLab Laboratories

- Four **Mandatory** MatLab Laboratories, Mondays from 6:00PM to 8:50PM [10%]
  - Lab 1: Introduction to MatLab, Week 3 (May 25)
  - Lab 2: Graphing and Contour Plotting,Week 5 (June 8)
  - Lab 3: Using the Symbolic Toolbox, Week 6 (June 15)
  - Lab 4: Triple Integration and its applications, Week 9 (July 13)
- For all labs, you will be coding and completing a worksheet.
- Check the Assessment and Activities in the Course Information Module in LEARN for all deadlines and assessment weights listed per week.

Missed Laboratories, Deadlines and Grace Days

- Missed MatLab Laboratories receive ZERO points for the corresponding Lab Worksheet (NO MAKE-UPS PERMITTED)
- Submit all code files and completed worksheets to the designated Dropbox by the stated deadline
  - A 25% penalty per working day (or part of the day) is applied to worksheets submitted after the deadline
- You are allotted a total of 4 Grace Days for the entire term
  - Grace days are counted for working days only
  - Once all grace days are used, a 25% penalty per working day will be applied to late submissions

i-Clicker Cloud

- i-Clicker Cloud will be used in our course during all Instructor lectures and tutorials
  - Allows you to show me what you understand without having to raise your hand and identify yourself
  - Allows you to review on the spot your understanding of the material being covered
- A maximum bonus of 5% will be assigned for at least 75% of questions answered
- To register for the i-Clicker cloud polling system for MTE 203 Spring 2026, go to the i-Clicker Website.
  - Create a new account using your watiam-userid@uwaterloo.ca as your email address. Your first and last name must match what it is in Waterloo LEARN

Thank you and let our course begin!

May the force be with us …

UNIVERSITY OF WATERLOO